# Yiqun Zhou

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## Research Interests

My current research focuses on enabling robots to better understand the real world and interact with it reliably over long horizons. My interests span embodied AI and vision-and-language navigation, world models and predictive representations, 3D perception and SLAM.

## Education

2023 – present **Zhejiang University**, College of Computer Science and Technology
Ph.D. student, Computer Science and Technology · Intelligent Vehicle Research Center
Advised by Prof. Zhijie Pan

2019 – 2023 **Chongqing University**, College of Computer Science
B.Eng., Computer Science and Technology (Excellence Program) · CET-6: 580

## Publications and Patents

Under review

1. **Yiqun Zhou**, Xiao Liu, Yuechen Wu, Junjie An, Jiaxiang Zhou, Yizheng Huang. *RouteGuide: Fast VLN Adaptation from a Successful Video.* ACL ARR 2026.
2. **Yiqun Zhou**, Jiaxiang Zhou, Yizheng Huang, Xiao Liu, Yuechen Wu, Feixiang Jing. *Imagine, Interpret, Act: Giving Temporal Latents a Sense of Progress.* AAAI 2027.
3. **Yiqun Zhou**, Jiaxiang Zhou, Yizheng Huang, Ziming Zhao, Xuhong Zhang, Jianwei Yin. *TUTOR-Bench: A Target-Conditioned Utility Benchmark for Cross-Task Robot Data Reuse.* Submitted to KDD 2027 Datasets and Benchmarks Track. Code.
4. **Yiqun Zhou**, Jiaxiang Zhou, Yuhao Chen, Fanqi Kong, Yu Peng, Yizheng Huang, Ziming Zhao, Xuhong Zhang, Jianwei Yin. *Spotter: Budgeting Agent Time and Camera Bytes for VLA Harnesses.* Submitted to ACM MobiCom 2027.
5. **Yiqun Zhou**, Jiaxiang Zhou, Yuhao Chen, Fanqi Kong, Yu Peng, Yizheng Huang, Ziming Zhao, Xuhong Zhang, Jianwei Yin. *TopoMoE: Topology-Aware Expert Routing for Mixture-of-Experts Inference over Mobile Edge Networks.* Submitted to ACM MobiCom 2027.

Published

6. **Yiqun Zhou**, Tengfei Hu, Xiao Zhou, Pan Lv, Zhijie Pan, Hong Li, Guoqing Yang. *CurbMapping: Reducing Blind Spots for Low-cost Autonomous Sweepers.* CAC 2025.
7. **Yiqun Zhou**, Huimin Zhao, Giandomenico Caruso. *Dedicated Lanes for Intelligent and Connected Vehicles in Mixed Traffic.* CIVS 2025.
8. Shengpu Zhang, **Yiqun Zhou**, Pan Lv, Guoqing Yang, Hong Li, Zhijie Pan. *Loosely Coupled LiDAR-Inertial-Encoder SLAM for Large-Scale Environments with Limited FoV.* CAC 2025.
9. Chao Liu, Runfeng Cai, **Yiqun Zhou**, Xin Chen, Haibo Hu, Meng Yan. *Understanding the Implementation Issues When Using Deep Learning Frameworks.* Information and Software Technology, 166, 2024. CCF-B
10. Guanyu Chen, **Yiqun Zhou**, Guoqing Yang, Hong Li, Pan Lv*. *SmartKit: User-Friendly Robot with Multiple Operating Systems.* IROS 2024. CCF-C
11. Lan Gao, **Yiqun Zhou**, Xin Chen, Runfeng Cai, Guo Chen, Chaojie Li. *Privacy-Preserving Dynamic Average Consensus via Random Number Perturbation.* IEEE Trans. on Circuits and Systems, 2022.

Patents Four invention patents (1 granted, 2 under substantive examination, 1 filed): HD map construction for unmanned vehicles in large outdoor scenes; LiDAR blind-spot obstacle avoidance for low-speed outdoor sweepers; behaviour planning on structured pavement; single-lane vehicle routing under dynamic demand.

## Research Projects

**Key technologies for vehicle-control operating systems in intelligent connected vehicles**
NSFC Joint Fund for Regional Innovation and Development · Led by Beihang University · Zhejiang University as collaborating institution · mid-term review March 2025
I contributed to the implementation of the SmartKit mixed-criticality system and completed ROS fault-recovery validation on an unmanned vehicle platform. The system uses virtualization for parallel execution and safety isolation across multiple operating systems, with a real-time OS monitoring the Linux guest and switching to a preloaded backup VM on failure. The work was published at IROS 2024; I prepared the mid-term review materials and presented as the Zhejiang University representative.

**Automated test and verification toolchain for automotive ECU software**
National Key R&D Program of China, "New Energy Vehicles" key special project · Led by Beihang University · Zhejiang University as participating institution
I led the ZJU student team in system development, designed the ECU inspection platform and ASAM protocol layer, and integrated HIL rigs for automated test-case generation and execution. I drafted the mid-term and final defense materials, organized expert and final acceptance meetings, and led the acceptance demonstration.

## Engineering Projects

**Deploying VLA policies and vision-language navigation on real robots**
Anker Innovations · Internship · 2025 – present
I built a physical-robot data-collection and deployment environment with a PICO headset teleoperation demonstration system and a localization module. I aligned images, joint states and issued commands for each trajectory, completed cloth-folding and related manipulation tasks with $\pi_{0.5}$ on Piper and Franka arms, deployed manipulation policies on a Unitree G1, and implemented natural-language-driven vision-language navigation on a Unitree Go2.

**Techinao RoboNeo — outdoor autonomous cleaning vehicles**
Wuxi Techinao Intelligent Technology (founded by our team) · techinao.com · Team lead · 2022 – present
I built the vehicle software stack from zero, spanning embedded control, sensor selection and integration for cameras and LiDAR (Mid-360, RTK), and the mapping, localization, navigation and planning modules. I deployed a MoGe-2-based monocular depth estimation system on the vehicle, and led the team through several vehicle generations into production. The RoboNeo Max / Pro / C99 line now runs daily in scenic areas, shopping malls and factory warehouses.

**Cloth Cart Vision — tracking carts in dyeing workshops**
Shaoxing Keqiao District Textile Industry "Vanguard and Leading Geese" Program · Embodied foundation models · Project page
The vision-based system recognizes cloth-cart IDs and load states, supports location lookup, stalled-cart alerts, trajectory history, order maps and empty-cart search, and synchronizes cart-change events with ERP / MES for automatic work-order-level segmentation of energy, process and production data. The system has since been productized and deployed in more than ten factories.

**Autonomous loader and sweeper at Lanshan Port, Shandong**
Shandong Port Group project · 2024
Mapping and localization at a working bulk-cargo port, deploying an unmanned tracked loader to clean along a three-kilometre spoil conveyor. Structural features are sparse, dust and vibration are constant, operation cannot be interrupted, and the three-kilometre run makes trajectory drift the dominant error source, so long-run stability matters more than instantaneous accuracy. I built the map offline with FAST-LIO, added a keyframe strategy to suppress drift, and used NDT registration for global relocalization so that the system recovers its pose quickly at start-up and after any interruption; deployed and tuned on site.

**Tianmen Mountain autonomous vehicle challenge — vehicle control**
2025
Responsible for vehicle control, adapting and tuning openpilot's lateral and longitudinal control on a real car. The course is the Tianmen Mountain summit road: 99 hairpin turns and 1,100 m of vertical drop, with intermittent satellite reception and limited visibility.

## Research Capabilities

Perception — I carry mapping, relocalization and calibration through the full pipeline in large real environments and keep them stable over long runs where structural features are sparse and interference is heavy — LiDAR-inertial odometry and NDT registration, point cloud semantic segmentation, multi-view geometry and camera calibration, RTK positioning and multi-sensor fusion.

Policies — I run the full VLA pipeline — data collection, training, inference and deployment on a real robot — and am familiar with mainstream VLA architectures and the action-chunking mechanism, as well as the design and evaluation of world models. I also know the mainstream computer-vision algorithms and am comfortable with training and deployment pipelines on the common compute platforms; related work includes closed-loop VLN evaluation, reinforcement learning, and VLM fine-tuning and distillation.

Systems — I take systems end to end, from sensor and algorithm integration to whole-machine deployment. I am familiar with ROS 1 / ROS 2 and its debugging toolchain, and with a range of robot platforms — Unitree Go2 and G1, Franka and Piper arms. I have delivered in the field on unmanned vehicles, manipulators and legged robots, with experience in edge-side inference optimization and real-time tuning.

## Awards

2024 — Outstanding Student, Zhejiang University · Head of Academic Affairs, Ph.D. Student Union

2022 — **National Scholarship**, Ministry of Education · Huawei "Rising Star" Scholarship · Third Prize, Huawei ICT Competition (National) · Third Prize, National Embedded Chip Design Competition · Outstanding Completion, National Undergraduate Innovation Program

2020 – 2022 — Top-10 Youth League Member and Outstanding Student, Chongqing University